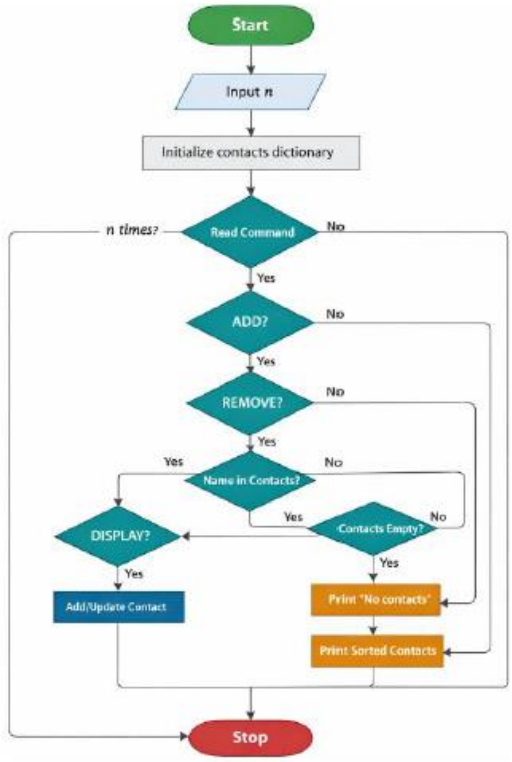
12.1.1 Phone Number Database Management

Algorithm:-

1. Start
2. Input number of operations `n`
3. Create an empty dictionary `contacts`
4. Repeat `n` times:
    - Read command input
    - If command is `"ADD"`:
        - Take `name` and `phone`
        - Add/update in `contacts`
    - Else if command is `"REMOVE"`:
        - Take `name`
        - If `name` exists in `contacts`, delete it
    - Else if command is `"DISPLAY"`:
        - If `contacts` is empty, print `"No contacts"`
        - Else print all contacts in sorted order of names
5. Stop

Flowchart



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12.1.1. Phone Number Database Management

Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.

The program should support three operations: adding a contact (if the name already exists, update the phone number), removing a contact (if the name does not exist, ignore the operation), and displaying all contacts in alphabetical order by name.

Input Format:

1. First line contains an integer n representing the number of operations.
2. Each of the next n lines contains a single operation in one of the following formats:
   - "ADD <name> <phone_number>" where <name> is a string, <phone_number> is a string of 10 digits.
   - "REMOVE <name>" - name is a string.
   - "DISPLAY" - no additional parameters.

Output Format:

- For display operation, print all contacts in alphabetical order by name in the format:

```
<name>: <phone_number>
```

- If the database is empty during display operation, print: "No contacts".
- Do not print anything for add and remove operations.

Constraints:

- Name contains only alphabetic characters (no spaces). And it is case sensitive.

phoneDat...

```python
n=int(input())
ph={}
for i in range(n):
    cmd = input().split()
    if cmd[0] == "ADD":
        ph[cmd[1]]= cmd[2]
    elif cmd[0] == "REMOVE":
        ph.pop(cmd[1], None)
    elif cmd[0] == "DISPLAY":
        if len(ph) == 0:
            print("No contacts")
```

Average time	Maximum time
0.024 s	0.032 s
23.75 ms	32.00 ms

5 out of 5 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 32 ms Debug

Expected output	Actual output
4	4
ADD Alice 9876543210	ADD Alice 9876543210
ADD Bob 8765432109	ADD Bob 8765432109
ADD Charlie 7777777777	ADD Charlie 7777777777
DISPLAY	DISPLAY
Alice: 9876543210	Alice: 9876543210

Terminal   Test cases

Prev   Reset   Submit   Next